

How can slip stick friction be maximized between finger and computer mice

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1 Introduction

Friction is a topic that has intrigued people for millenia, and pioneered to this very day. Think honestly, what causes friction? Small microscopic asperities? Then how does perfectly smooth rubber still present friction? and why would jagged ice be so slippery? Is water the issue? Then why can you tread on water? From scientists, control engineers, and frankly, scholars that experienced its' effects in other forms, are all trying to understand this confounding concept.

In particular, dry friction sometimes results in behaviours such as doors creaking, violin vibrations, etc. The oscillating change between kinetic and static friction in a motion. Take for instance a mass pulled by a spring moving at constant velocity v_p as illustrated in figure 1. In the beginning, the mass is stationary with an increasing applied force from the spring towards the right with static friction of equal magnitude pulling

the opposite direction, once the elastic force exceeds the static friction force, accelerating the mass forward before stopping and looping back at to the beginning with elastic force being lower than the static friction.

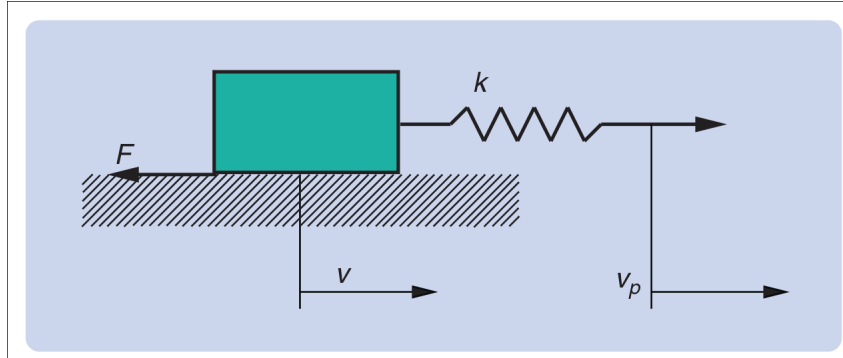
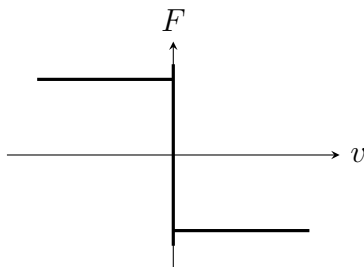


Figure 1: Mass being pulled by a spring at constant velocity [4]

It is a topic fixated by engineers in trying to reduce, common solutions include adding grease, oil, etc. However, in the realm of competitive Minecraft, there exists many instances where the desirable effect of clicking fast can be applied by exploitation of slip stick atop the mouse. This is referred to in the community as “drag clicking”.

In this paper, we will investigate how different materials with different LuGre parameters can affect properties of slip stick friction between finger and surfaces. Such properties include cycle frequency, critical force to enter slip-stick oscillations, as well as other observations of interest.

2 Background



Our commonly conceived model of friction—the coulomb friction model—has several limitations. It suggests that static frictional force is proportional to the normal force, and that upon motion the frictional force becomes lower and whose magnitude is proportional to the normal force of the opposite direction of the velocity [4] illustrated in figure ??.

$$F_c = \mu_c F_N \quad (1)$$

Common physics textbooks have this perceived as due to microscopic asperities interlocking with one another (Figure 2).

This, for instance explains the independence of frictional force on apparent surface area and how its rather dependent on normal force. this is because the normal force pushes more asperities in contact, therefore true microscopic contact area between asperities is much larger despite small appearant contact area. However, friction still includes much more confounding exceptions such as:

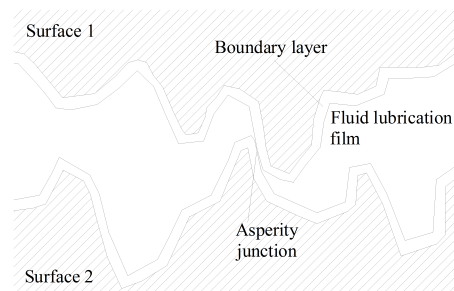


Figure 2: Illustration of Asperities [6]

1. The Stribeck Effect
2. Microscale pre-slip deformations
3. Dependence on past events
4. Ultimate friction aging

2.1 Friction Phenomena

2.1.1 Viscous Friction

Viscous friction introduces the role of velocity in that sliding friction increases with velocity. Though exact instantaneous velocity cannot be provided via discrete data points such as simply recording a video on drag clicking because two frames in a video only provides average velocity rather than true instantaneous velocity, and is an impractical path considering the velocity vector changes a lot during drag clicking, it becomes increasingly difficult to experimentally determine values with this setup, a treadmill setup could mitigate this since running it at a constant velocity ensures mutual understanding of the velocity. Sketched in figure 3

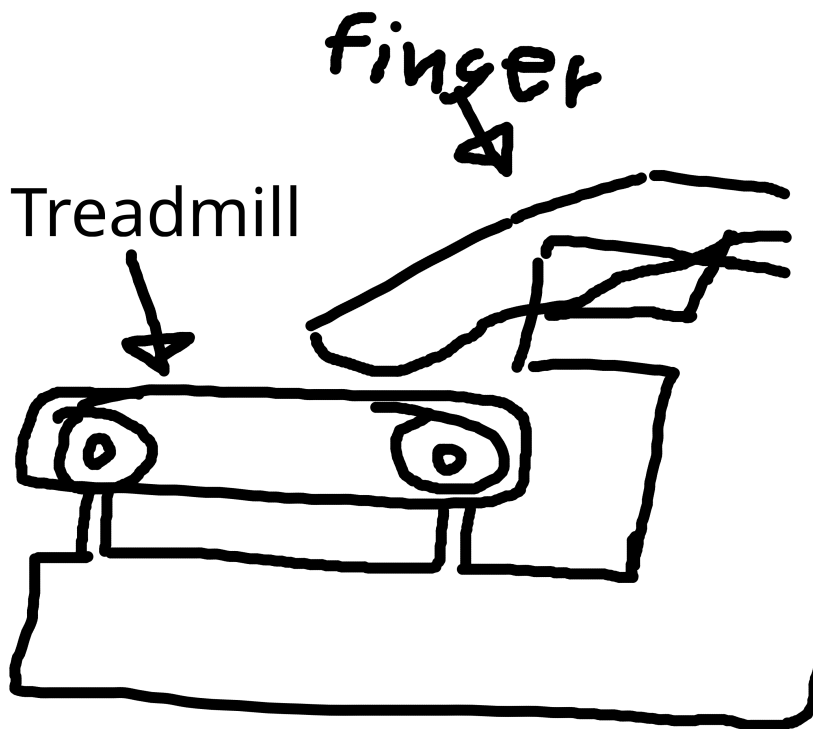


Figure 3: Sketch of a treadmill setup

2.1.2 Static Friction

Static friction describes the maximum frictional force that is reached at rest. When applied force is beyond this threshold, the friction shifts from static friction to kinetic

friction (lower than static friction). The process of this shift is often the exciter for slip stick friction.

2.1.3 The Stribeck Effect

The Stribeck Effect suggests that static friction does not instantaneously jump to kinetic/viscous friction but happens gradually. The rate of friction mode change depends much on material properties and in particular, we expect sharper jumps to exhibit superior drag-clicking capabilities. This means to comprehend this we cannot use coulomb friction's model as it does not consider friction as a function of velocity and is problematic because it allows the following interpretation:

Suppose the case where the applied force $F_A = 2N$ is less than maximum static friction $F_{S,max} = 4N$, the opposite force currently would be $2N$ and the net force $F_{net} = F_A - F_S = 0$ would result in no motion.

According to the coulomb's model, the instant at which the applied force $F_A > F_S$ the system will shift into kinetic friction. For the instance that $F_A = 4N$ increases to $F_A = 4.01N$, it shifts immediately to kinetic friction, which is lower than static friction. If kinetic friction $F_k = 2N$ then the net force increases from $F_{net} = F_A - F_{S,max} = 0$ for $F_A = 4N$ to $F_{net} = F_A - F_k = 4.01 - 2 = 2.01N$ for $F_A = 4.01N$ despite the tiny change in applied force. However, in real life, it is not possible because some guys did a research and tried to immediately change the force from 4.01 to $2.01N$ during the transition to kinetic friction and was able to achieve a net force of $0.01N$ which suggests the transition is not an immediate process.

The reason why a steeper slope would be more desired for drag clicking is contrasted between the two cases below: If the jump between static friction and kinetic friction were more steep (such as the optimal case of coulomb friction), then the increase in net force would be more drastic and this drastic change allows the finger to experience instant acceleration which leads to the illusion of it clicking and having swung forward to the next cycle. However, if the transition were less instant, for example when applied force was $4.01N$ and after 10 microseconds the finger has accelerated to a velocity of $50cm/s$ and at that velocity the friction is $3N$, we would have a net force of $1.01N$ rather than the optimal case described in the coulomb friction (of which more closely resembles the desired steep curve).

2.2 State Space Equations

Consider the condition at which the same velocity and normal force results in different frictional forces [2]. This is because of the dependence of friction on previous states.

While the standard textbook definition for friction stem from rigid body microscopic asperities, the Dahl Model instead visualizes two bodies making contact from elastic bristles scattered over the contacting surface[2]. Illustrated in figure 4.

For instance, if the velocity and normal force be the same, having the bristles against the direction of motion would result in larger frictional force than if the bristles were aligned with the direction of motion.

The amount of displacement is represented by the state variable $z = \frac{F}{\sigma_0}$, where σ_0 is the stiffness of the bristles and, F is the frictional force. This can be perceived as analogous to the elastic force equation in $F_H = -kx$.

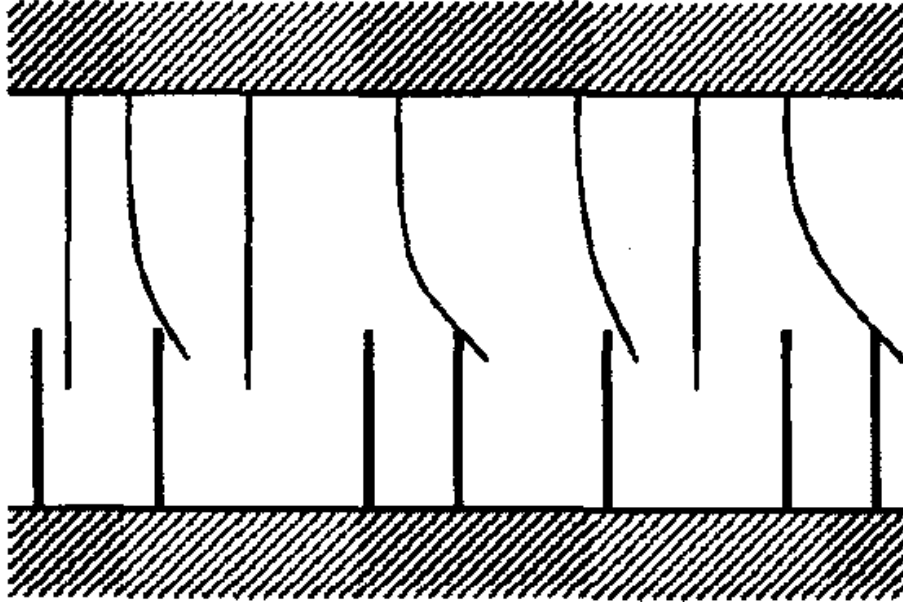


Figure 4: Depiction of Dahl Model's Bristles [2]

To coherently account for behaviour involving state changes as a result of internal states, we must shift to dynamic models that uses state space equations as it accounts for hysteresis behaviour[5].

In state space equations, the derivative of state variable is defined as a function of state variables, such as the LuGre model:

$$\dot{z} = v - h(v)z \quad (2)$$

which suggests that the rate of change of the bristle deflection is the difference between the velocity and the previous bristle deflection with regards to a function of velocity. Plugging in all known values and integrating this function over a tiny unit of time gives us the state at the next instant of the bristle deflection z [5]. Now that we have obtained both $\dot{z}$ and z with a known v we can plug into the second equation of the lugre model

$$F = \sigma_0 z + \sigma_1 \dot{z} + f(v) \quad (3)$$

As a result it gives us a prediction of the state successively to give a sequential account of the predicted friction. And looking at the relationship between variables in a state space equation can describe the stability of a system [5], which is particularly useful in stick-slip motion.

2.3 The Dahl Model

Since the Dahl Model was built on the concept of elastic bristles, the foundation was based on the stress strain curve:

$$F = F_c \left(1 - e^{-\frac{\sigma_0 |x|}{F_c}} \right) \text{sgn} \left(\frac{dx}{dt} \right) \quad (4)$$

- F_c is the coulomb friction, represented here as the maximum friction boundry

- x is the displacement between surfaces similar to that in $F_H = -kx$, though, in this model it focuses primarily on microscopic pre-displacement. As x increases, $e^{-\frac{\sigma_0|x|}{F_c}}$ decreases, which makes sense because by applying more force, predisplacement x increases which results in more opposing frictional force.
- σ_0 is as aforementioned the stiffness of the bristles, analogous to the spring constant k in the Physics Formula Booklet. Similar to x , if σ_0 is larger, it means that bristles are more stiff, and makes $e^{-\frac{\sigma_0|x|}{F_c}}$ easier to approach 0, making it easier to reach maximum friction.

Since state space equations are more concerned with the change in state variables, and are defined as derivatives, we must find the derivative of F with respect to x using chain rule in order to reveal how the force changes as the predisplacement x changes:

$$\frac{dF}{dx} = \cancel{F_c} \cdot e^{-\frac{\sigma_0|x|}{F_c}} \cdot \cancel{F_c} \sigma_0 \operatorname{sgn}\left(\frac{dx}{dt}\right) \operatorname{sgn}(x) \quad (5)$$

$$= \sigma_0 e^{-\frac{\sigma_0|x|}{F_c}} \operatorname{sgn}\left(\frac{dx}{dt}\right) \operatorname{sgn}(x) \quad (6)$$

and to put the actual current force state back into this derivative function (since state space equations depend on previous states), we rearrange eq (1) to isolate $e^{-\frac{\sigma_0|x|}{F_c}}$ like such:

$$F = F_c \left(1 - e^{-\frac{\sigma_0|x|}{F_c}}\right) \operatorname{sgn}\left(\frac{dx}{dt}\right) \quad (7)$$

$$\frac{F}{F_c} \operatorname{sgn}\left(\frac{dx}{dt}\right) = 1 - e^{-\frac{\sigma_0|x|}{F_c}} \quad (8)$$

$$e^{-\frac{\sigma_0|x|}{F_c}} = 1 - \frac{F}{F_c} \operatorname{sgn}\left(\frac{dx}{dt}\right) \quad (9)$$

and putting it back in the differential equation we get:

$$\frac{dF}{dx} = \sigma_0 \left(1 - \frac{F}{F_c}\right) \cancel{\operatorname{sgn}\left(\frac{dx}{dt}\right)^2} \operatorname{sgn}(x) \quad (10)$$

$$= \sigma_0 \left(1 - \frac{F}{F_c}\right) \operatorname{sgn}(x) \quad (11)$$

The Dahl model's equation shares similar properties of an exponential function take of the stress-strain curve (depicted in figure 5), most notably, the point at which it shifts from an elastic spring to continual bristle movement

Most importantly the Dahl Model introduces $z = \frac{F}{\sigma_0}$ as a state variable that represents the "average bristle deflection" [46]. This is very similar to Hooke's law. rearranging, we get:

$$F = \sigma_0 z \quad (12)$$

similarly, in Hooke's law presented in the Physics formula booklet:

$$F = kx \quad (13)$$

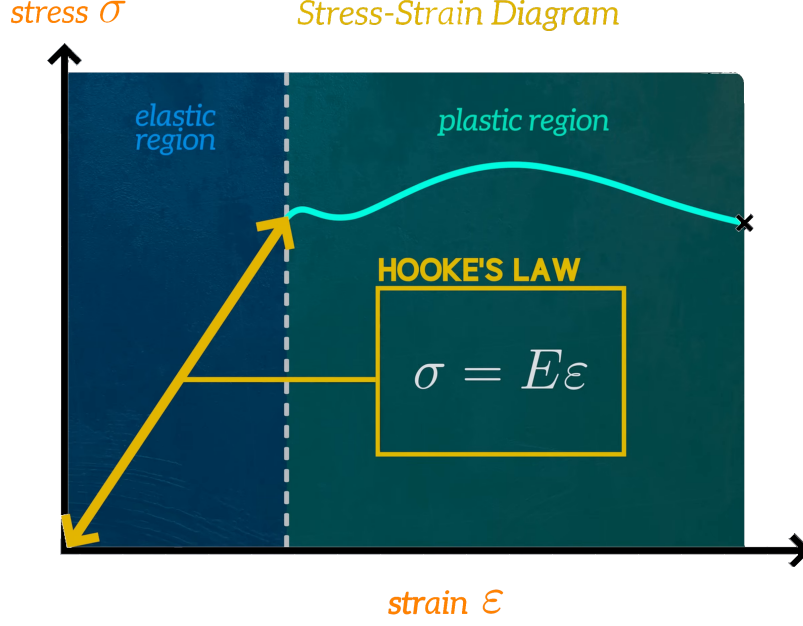


Figure 5: An annotated stress-strain curve[3]

Differentiating $z = \frac{1}{\sigma_0} \cdot F$ to find the state equation:

$$\frac{dz}{dt} = \frac{1}{\sigma_0} \cdot \frac{dF}{dx} \cdot \frac{dx}{dt} \quad (14)$$

$$= \frac{1}{\sigma_0} \cdot \frac{dF}{dx} \cdot v \quad (15)$$

$$(16)$$

and with $\frac{dF}{dx} = \sigma_0 \left(1 - \frac{F}{F_c} \text{sgn}(v)\right)$

$$= \frac{1}{\sigma_0} \cdot \sigma_0 \left(1 - \frac{F}{F_c} \text{sgn}(v)\right) v \quad (17)$$

$$= \left(1 - \frac{F}{F_c} \text{sgn}(v)\right) v \quad (18)$$

$$(19)$$

now with $F = \sigma_0 z$ and $\text{sgn}(v)v = |v|$ we arrive at the definitive Dahl's model

$$= \left(1 - \frac{\sigma_0 z}{F_c} \text{sgn}(v)\right) v \quad (20)$$

$$\frac{dz}{dt} = v - \frac{\sigma_0 z}{F_c} |v| \quad (21)$$

[4]

2.4 Lugre Model

The Lugre model takes in consideration that coulomb friction is not constant but rather dependent of velocity, thereby replacing it with $g(v) = F_c + (F_s - F_c) e^{-|\frac{v}{v_s}|^\alpha}$ that considers the rate of change from F_s to F_c to be non-instantaneous as shown in figure 6

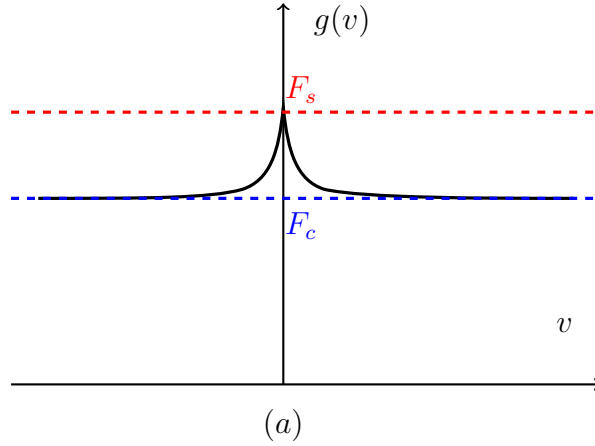


Figure 6: Graph of $g(v)$ with relative F_s and F_c coordinates

to create the following differential equation:

3 Experiment

3.1 Drag clicking specific methods

There exists several known methods in the community to increase the accessibility of drag clicking on a mouse. With any youtube search you will find it populated with applying "tape" to your mouse, along with sharpie markers applied to that, then the mention of keeping your hands dry. In private I have discovered several new methods to increase friction, along with additive methods, I have discovered subtractive methods. Additive methods include rubbing crushed cello rosin dust onto the surface of the mouse with your finger, it was meant to adjunct slip stick properties of stringed instruments, though the effect is universal. Subtractive include rubbing the surface of your mouse with the bottom of your mousepad firmly, or with the leaf of a well hydrated plant. We will measure the rate at which static friction equalizes to coulomb friction with respect to time

3.2 Research Hypothesis and Variables

The difference between Coulomb friction and static friction changes depending on material surface. This investigation aims to determine how material surface affects the difference between coulomb and static coefficient of friction. It is hypothesized that Cello rosin will have the largest change due to its sticky nature, and clear PC plastic is expected with the least coefficient of friction as the surface is factory new surface polished The independent variables are clear pc, cello rosin, and sharpie while the independent variable is the difference between static and coulomb coefficient of friction.

3.3 Method

The static and kinetic friction will be found by measuring steady friction force at various velocities [4]. The following diagram shows the setup:

A finger sits on a scale while taped to a string horizontally, which passes through a pulley upwards connecting to force gauge with a force exerted by a motor (Vex V5 Smart

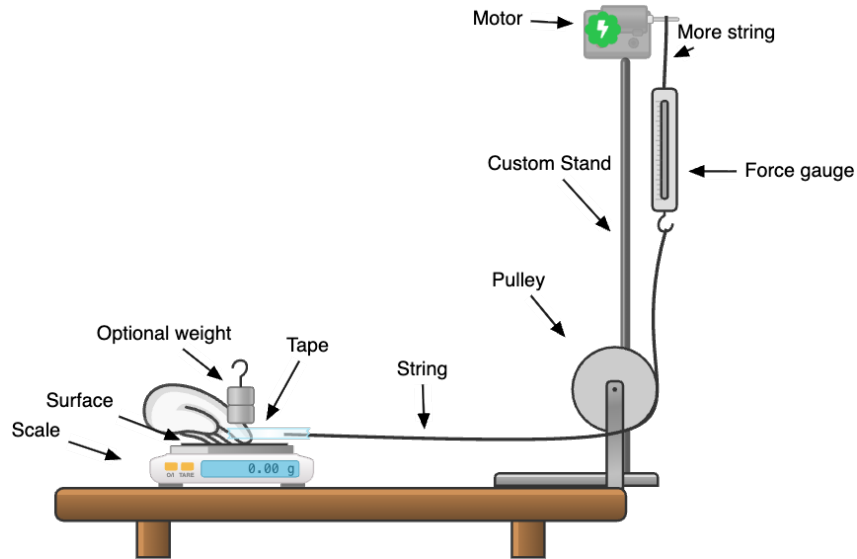


Figure 7: The experimental setup for measuring Coulomb and static friction

Motor 11W). The reason it is upwards is because if it were downwards, the weight of the force gauge would inherently exert a force due to gravity despite no applied force from the motor pulling it down and still displaying zero.

The experimental procedure upon procuring this setup is:

1. Clean the surface with alcohol, or reapply rosin, sharpie will not be reapplied upon initial application
2. Clean the finger with alcohol, or reapply rosin, again sharpie is neglected
3. Tare the scale and while at null force, tare the force gauge
4. Insert the string into the pulley
5. Place finger onto the scale
6. While recording slow motion, ensure visibility:
 - (a) Finger state (slip/stick)
 - (b) Force gauge value
 - (c) Scale value
7. Start the motor
8. repeat steps 1-8 3 trials
9. repeat for different materials

4 Results and Discussion

Table 1: PC Board

Trial 1	F_n	F_{fr}
	252	70
	231.5	
	210.5	
	215	
	228.5	
	209	
	199	
	238	
	243	
	248	
	259	
	269	
	252.5	
	250	
	262.5	
	252.5	
	269.5	
Average	241.1470588	70
$\mu_{fr} = \frac{F_{fr}}{F_N}$	0.290279302	

Trial 2	F_n	F_{fr}
	322	245
	248	
	230	
	273	
	267.5	
Average	268.1	245
$\mu_{fr} = \frac{F_{fr}}{F_N}$	0.91383812	

Trial 3	F_n	F_{fr}
	227.5	150
	178	
	210.5	
	199.8	
	179.2	
	194.3	
	222.5	
	225	
Average	204.6	150
$\mu_{fr} = \frac{F_{fr}}{F_N}$	0.73313783	

Table 2: Cello Rosin

Trial 1	F_n	F_{fr}
	216.5	120
	165.2	
	172.7	
	180	
	191.6	
	181.3	
	214	
	265	
	271.5	
	239	
	245	
	264.5	
	243.5	
	253.5	
	233.5	
	225.5	
	249	
	251.5	
	263.5	
	294	
	300	
	300	
	295	
	304.5	
	302	
	300.5	
	296	
	304	
	302	
	304	
	317	
	317	
	320	

Trial 1	F_n	F_{fr}
	337.5	
	349	
	347.5	
	349.5	
	362.5	
	366	
	363.5	
	361	
	352.5	
	370.5	
	372	
	372	
	370.5	
	364	
Average	289.793617	120
$\mu_{fr} = \frac{F_{fr}}{F_N}$	0.414087795	
Trial 2	F_n	F_{fr}
	444	120
	444	
	428.5	
	428.5	
	428.5	
	423	
	423	
	400.5	
	400.5	
	383.5	
	376.5	
	278.5	
	286.5	

Trial 2	F_n	F_{fr}
	311.5	
	322.5	
	322.5	
	316.5	
	316.5	
	329.5	
	329.5	
	344	
	458	
	458	
	458	
	462.5	
	462.5	
	462.5	
	456.5	
	456.5	
	456.5	
	450	
	450	
	442	
	442	
	444	
	439	
	462.5	
	462.5	
	501.5	
	517.5	
	517.5	
	514	
	519.5	
	516.5	
	516.5	
	507	

4.1 Observations

The electrical tape was unable to enter kinetic friction, this is likely due to the fact that the α in the lugre model is too large, meaning the transition to coulomb friction was too rapid (i.e. any increase in velocity resulted in a rapid decrease of frictional force to keep from sliding). It could be possible for the finger to indeed be in motion, except that the velocity is close to zero, explaining the seeming impossibility of reaching visible coulomb friction. Since the tape is viscoelastic, it has a strong elastic spring property, that like described in figure 5, the resultant frictional force is directly proportional to displacement, of which is a result of applied force, which explains the difficulty in achieving kinetic friction, and once the applied force reaches the breaking point, it slips. the addition of the elasticity of the finger introduces additional difficulty in reaching kinetic coulomb friction since elastic on elastic contact gives extra opportunity for the reaction force to reflect the applied force.

For finger systems, it could be suggested that a modified lugre model should be used in which the σ_0 in $z = \frac{F}{\sigma_0}$ reflects the spring constant of a finger rather than microscopic bristles.

The trials of rosin exhibited slip stick of short slip distance before immediately sticking and slipping again. This could be a result of the material being non-viscoelastic, as such, the friction did not have the double spring effect, otherwise it would have doubled the spring of finger and material's displacement before finally being able to slip. Instead, it immediately jumped to coulomb in the case of sufficient applied force. This would be the result of a high α in lugre model's variation of coulomb friction as a result of velocity described in $g(v)$. And this could be a result of particulate friction that has not been looked into in this research, but the stickiness of rosin dust on the finger could be responsible for replacing the skin sebum from contacting the surface of the rosin on the pc board, and instead change into rosin on rosin contact.

4.2 Limitations of the data

The force sensor used was for measuring masses and would stop updating when no change is detected. This significantly impacted the accuracy of the data. Further improvement in methodology could involve high frequency force meter and scale that directly output into computer for high frequency data collection, in particular a computer with high monitor refreshrate and subsequently a better slow motion camera can capture much more details entailing the phases of stick slip behavior. Another limitation is the subject's muscle contraction when resetting, it is possible that force is applied prior to place the finger at the end for the string to not fall out of the pulley. It is also possible for the subject to unintentionally produce opposite force in the same direction of the frictional force to maximize slip stick behavior. Perhaps a future experiment with a moving surface rather than a moving finger would be used instead similar to [1]

Trial 2	F_n	F_{fr}
	505	
	505	
	502	
	506.5	
	506.5	
	490.5	
Average	434.3557692	120
$\mu_{fr} = \frac{F_{fr}}{F_N}$	0.276271224	
Trial 3	F_n	F_{fr}
	246.5	215
	246.5	
	321	
	321	
	331	
	331	
	331	
	312.5	
	312.5	
	271	
	297	
	349.5	
	365.5	
	365.5	
	385	
	377	
	377	
	352.5	
	355.5	
	374	
	380	
	380	
	391	
	391	
	402.5	
	388.5	
	375	
	380.5	
	383.5	
	369	
	351	
	340.5	
	350	
	350	
	344	

Trial 3	F_n	F_{fr}
	320.5	
	320.5	
	326	
	326	
	327.5	
	327.5	
	344.5	
	346	
	346	
	365	
	389.5	
	395	
	395	
	395	
	397	
	397	
	394.5	
	394.5	
	399.5	
	399.5	
	412	
	414.5	
	400.5	
	400.5	
	381.5	
	377.5	
	384.5	
	385.5	
	385.5	
	385.5	
	380	
	380	
	385	
	385	
	385	
Average	362.1071429	215
$\mu_{fr} = \frac{F_{fr}}{F_N}$	0.593746918	

Table 3: Sharpie

Trial 1	F_N	F_{fr}	μ_{fr}
	218	100	
	198.4		
	220		
	242.5		
	210.5		
Average	217.88	100	0.458968239
Trial 2	F_N	F_{fr}	μ_{fr}
	179.1	515	2.875488554
	148.3	535	3.607552259
	178.5	475	2.661064426
	263.5	635	2.409867173
	309.5	610	1.97092084
	269	410	1.524163569
Average			2.508176137
Trial 3	F_N	F_{fr}	μ_{fr}
	187.1	210	1.122394441
	218	265	1.21559633
	234	405	1.730769231
	231	465	2.012987013
	241	505	2.095435685
	226	535	2.367256637
	215	550	2.558139535
	216.5	555	2.563510393
	262	580	2.213740458
	313	555	1.773162939
	316.5	590	1.864139021
	322.5	620	1.92248062
	333.5	760	2.27886057
	403.5	715	1.771995043
	402.5	165	0.409937888
Average			1.860027054

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